

Bidirectional Electrothermal Inchworm Actuator

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Abstract—Traditional MEMS actuating methods including electrostatic and electrothermal actuation are widely used throughout this field due to their simplicity, but their range of displacements is relatively limited and power efficiency is suboptimal. We propose a bidirectional, multi-staged electrothermal inchworm actuator controlled by a CMOS digital sequencing architecture. The system combines a 180 nm CMOS control chip with an off-chip heater-driving stage that delivers approximately 100 mA per active actuator using a 1.8V control signal and external 7 V heater supply. By coordinating multiple electrothermal actuator stages and a save-state holding mechanism, the design targets up to 50 micrometers of movement left or right from the fabricated position, a limit which scales drastically with increasing die size. This method enables great displacement with very low steady state power usage, enabling many complex designs that were previously thought to be impossible. Upon fabricating and testing this MEMS device, we weren't able to reach this target specification, only producing ~3 microns of actuation, but we did prove the general operation principle viable for future projects.

I. INTRODUCTION

Background Information

MEMS have been driven using several major actuation methods, most commonly electrostatic or electrothermal actuators. Each actuation method offers distinct advantages but also important limitations. Electrostatic actuators typically require very low steady state power already because they do not rely on continuous current flow, but typical displacements are limited to very small numbers [1]. Electrothermal actuators provide better displacements, but they require a continuous current to hold their displacement, wasting a lot of energy in thermal heat loss to the environment [2]. From circuit perspective, this also creates a control challenge: the actuator must be driven in a precisely timed sequence while still minimizing on-chip power complexity. Therefore, our design combines digital CMOS control with staged MEMS actuation so that branch activation can be coordinated electrically while large displacement is achieved mechanically.

Proposed Actuation Method

Our goal is to make an actuator capable of producing large displacements without any continuous energy drain. This is achieved by using smaller electrothermal actuators to engage the inchworm structure sequentially, so that the central body is

advanced in multiple controlled stages. Our design principle is more clearly illustrated below in Figure 1. We will fabricate a central “inchworm” structure connected to the left die wall via a serpentine spring, allowing movement to the left and right. In the rightward motion case, the first actuator pair pinches into the inchworm providing an initial displacement and stretching the spring. The next stage will now pinch down and pull the inchworm even further - once it is fully engaged, the first stage can release. This process can be repeated periodically to achieve very high displacements. To realize this actuation sequence electrically, the microsystem includes a CMOS control chip that generates the digital motion sequence and heater activation commands. In particular, a finite-state controller and pair decoder are used to determine which actuator branches should be turned on at each step of the inchworm sequence.

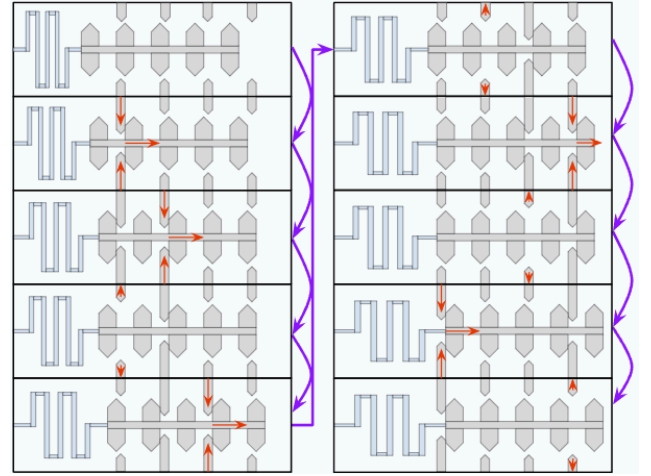


Fig. 1. This is a simplified form of our device with each stage working together to push the central mass to the right

This process shown in Figure 1 can be repeated as long as sufficient tooth count and spring compliance are available. The same concept can also be applied in the opposite direction, enabling bidirectional inchworm motion. A save-state branch is included so that the displacement can be retained mechanically when the system is unpowered. In the current design, however, the achievable displacement is still limited by the spring force. At the circuit level, this bidirectional operation is supported by a shared state sequence together with direction input, so that the same controller can drive either leftward or rightward motion without duplicating the motion logic. In addition, the save-state branch is included in the control architecture as a dedicated heater path that supports displacement retention within the staged actuation sequence.

Previous Attempts

Other groups have tried this in the past with only two actuator arms, and they have gotten impressive results, proving viability of unidirectional save state actuation. We are going to try and improve upon their design, all while having access to a less capable fabrication process [3][4]. Their implementations only use a limited amount of actuators, while

we are going to try and stack actuators to get more movement. From the control side, our approach also differs in that the branch-selection logic is handled by a dedicated CMOS sequencing architecture rather than relying only on the mechanical arrangement itself. This allows the staged pattern to be implemented in a more modular way and makes the heater activation sequence easier to define and verify.

Potential Applications

Our inchworm that we developed in this project isn't designed for one application. Instead it is designed as a proof of concept for any applications that require precise motion over a large distance. As the design has our multi-stage actuation it can achieve much larger displacement than many conventional MEMS actuators while still maintaining fine positioning capability. In particular this project targets lateral motion on the order of tens of microns, with powered resolution in the low micron range, making the device especially promising for microsystems where compact size, repeatable motion, and controllable displacement are all needed. Its ability to preserve position with little or no steady state power also makes it attractive for systems that must hold a mechanical state for extended periods without continuous energy consumption.

A major application of this is where you need to adjust small but precise distances. If you were positioning a lens, this device could prove useful - we would be able to line up the lens perfectly and get very consistent results.

Another possible application sees the placement of one of these dies on a microrobot to allow for locomotion. Imagine sitting in a sled atop a flat sheet of snow. You can move forward with a "scooting" motion without even touching the snow, all you have to do is slowly lean forward and quickly lean back. The principle here is the same - by taking advantage of the material difference between static and kinetic friction, the microrobot could charge the spring and release it all at once to propel it forward.

One further application would involve a microgripper with varying size grippers. Since the device allows save-state operation with 16 micron resolution, our initial design already allows for gripping and holding onto multiple different sizes of objects. This may prove useful in catching and holding onto biological matter such as cells.

From the circuit perspective, these applications also require a reliable and compact control interface. The use of a low-voltage CMOS sequencing chip together with an off-chip heater-driving stage makes the system more practical for integration, since digital motion control can be preserved without placing a high-voltage, high-current power stage inside the chip.

Report Overview

This report is organized as follows. Section II presents the overall microsystem operation, including the system architecture, signal flow, and functional partition between the CMOS control chip, the off-chip heater driver, and the MEMS

actuator. Section III describes the design and simulation of the main circuit and MEMS subsystem, including the finite-state controller, pair decoder, off-chip heater driver, actuator structure, and spring design. Section IV summarizes the MEMS layout and process description. Finally, Section V concludes the report with a summary of the main design results and system-level specifications.

II. MICROSYSTEM OPERATION

Full Microsystem Block Diagram

We are designing an actuator, so the circuits must provide the correct electrical inputs to the MEMS structure in order to produce the desired mechanical behavior. In contrast to a sensor microsystem, where the MEMS device first responds to an external physical variable and the circuit processes the resulting signal, our system operates in the opposite direction: user commands are first interpreted by the control circuitry, and the MEMS device then responds accordingly. The overall microsystem signal path, from user input to realized mechanical output, is illustrated in Figure 2.

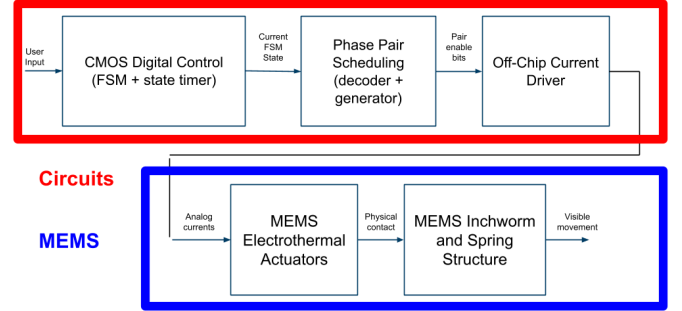


Fig. 2. Full microsystem block diagram, delineating circuit to MEMS pipeline. The user provides a control command, and the MEMS device reacts accordingly

The system starts with a user command that specifies whether the inchworm should move left, move right, hold its current position, or reset to the initial fabricated equilibrium position. The FSM portion of the circuit keeps track of the current motion state and determines the next desired state in the sequence. Each state corresponds to a different set of actuators, so accurate state progression is essential for correct motion in either direction.

The FSM state is then sent to the pair decoder, which determines which actuator branches should be turned on or kept off. Its five digital control outputs are applied to the off-chip current driver, which uses the external heater supply to energize the selected MEMS actuator branches. The resulting actuator motion and inchworm interactions can then be verified by optical observation with a microscope.

MEMS Overview

This subsection summarizes the final MEMS layout for die one and identifies major structures. The three major structures that most directly affect our final performance are the spring (green), the inchworm (red), and the actuators (thin blue lines), illustrated below in Figure. 3.

This device layout works the same way as shown in Figure 1, but instead of all the actuators being on the same level they are stacked at different vertical levels. This is because each actuator needs a large horizontal area, so we need to expand into the available vertical space, and add extra arms to the inchworm structure to reach into these contact points. The same principle applies where multiple actuator stages work together, moving the inchworm body in controlled increments.

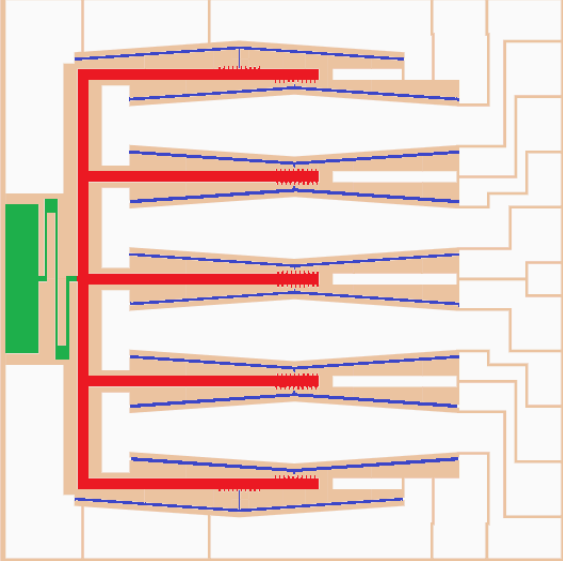


Fig. 3. Specific operation principle as implemented in our primary design

The save-state actuators act as the latching elements of the device. Unlike the other actuator pairs, which are used to advance the inchworm from one tooth position to the next, the save-state actuator is designed to sit between the teeth of the central inchworm structure when engaged. During motion, the save-state actuator is first pulled out of the way so the inchworm can translate. After the moving body reaches the next position, the save-state actuator is re-engaged so that its finger again rests between the teeth, mechanically holding the new position. It also works as a normal tooth and helps push the stages. You can recognize them at the top and bottom of Figure 3 because of their reversed polarity.

Our primary design allows targets very precise movement specifications (Table 1), generally dictated by our fabrication limitations. This includes 13 microns of movement in either direction to reach the first save-state actuator on each side, maximum displacements of 50 microns due to physical constraints, and displacement resolutions of 3.2 microns when powered, vs 16 microns when unpowered. All of these parameters were significantly influenced by process rules given to us, and we used minimum design features very frequently here. For the actuators themselves, we imposed a maximum input current of 100 mA as a rule specific to our design. This helped limit the risk of overheating and blowing up our device, and also worked in tandem with the 200 mA pad aluminum pad current limit. Within these current requirements, we aim to produce enough displacement to generate physical contact between the actuators and the central

inchworm structure - a necessary detail for the functionality of our device.

Category	Specification	Expected performance
MEMS	Target Minimum Displacement	+/- 13 microns lateral movement
	Maximum Displacement	+/- 50 microns lateral movement
	Displacement Resolution (powered)	3.2 microns
	Displacement Resolution (unpowered)	16 microns
	Actuator Current	100 mA
	Max Power Required	3-6 W
	Temperature Reached	450° C

Table 1. MEMS Performance Specifications

Circuits Overview

The circuit portion of the microsystem provides the digital control for the bidirectional electrothermal inchworm actuator. The circuit subsystem serves as the electrical control interface of the microsystem. Its role is not to directly generate large heater power on chip, but to produce a low-voltage times activation sequence that determines which MEMS heater branches are turned on at each step of the inchworm. In this way, the sequencing function and the power-delivery function are intentionally separated. At the system level, the circuit is organized into three main functional blocks: a finite-state controller, a pair decoder, and an off-chip heater-driving interface. The finite-state controller generates the motion state sequence, while the pair decoder translates each state into heater branch activation signals. These low-voltage control signals are then applied to the off-chip high-voltage switching stage, which drives the necessary current into the MEMS heater branches and produces the physical actuation.

The signal flow of the system-level behavioral simulation and input-output waveforms are as follows. At the system interface level, the primary circuit inputs are CLK, direction, enable, and reset, while the primary outputs are five heater-control signals: SS_enable, Pair1_enable, Pair2_enable, Pair3_enable, and Pair4_enable. These outputs define which MEMS heater branches are active in each motion state and therefore determine the physical actuation path of the inchworm. The finite-state controller generates the discrete motion state FSM_state together with the global control conditions, and the pair decoder converts these inputs into five heater-control outputs, which are then applied to the external current driver

to actuate the corresponding MEMS heater branches. Figure 4 shows the circuit block diagram, and Figs. 5 and 6 show the corresponding top-level simulation results, including the heater-enable sequence and the resulting branch-current activation pattern.



Fig. 4. Top-level circuit block diagram

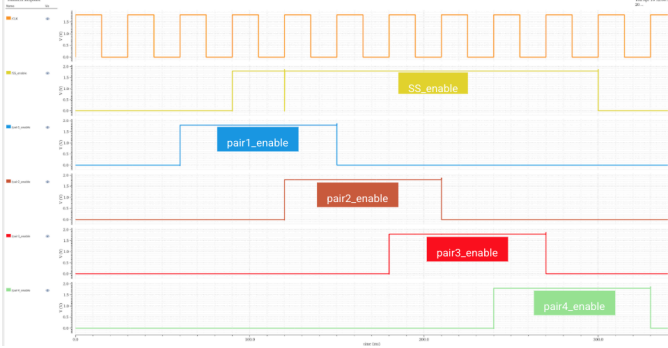


Fig. 5. Top-level simulation. The system generates five stages: SS_enable, pair1_enable, pair2_enable, pair3_enable, and pair4_enable, which sequentially move the central structure.

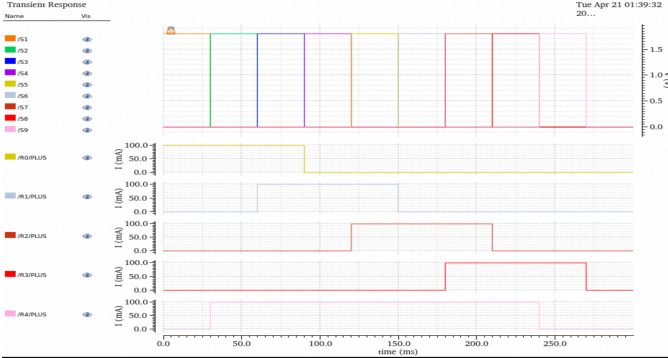


Fig. 6. Output heater branch currents in the system-level model. The x-axis denotes simulation time (ms), and the y-axis denotes branch current (A), showing the activation pattern of the MEMS heater branches during the inchworm sequence.

This partition defines the system interface between the low-voltage CMOS controller and the heater actuation path. The chip outputs only the branch-selection control signals, while the external driver uses a feedback-regulated heater-driving stage to supply the target current for the selected branch. In this way, the digital control path and the high-power actuation path remain electrically separated at the microsystem level. A key system-level challenge is interfacing low-voltage CMOS logic with the higher-voltage, higher-current requirements of the electrothermal MEMS heaters. The control path must generate the correct bidirectional activation sequence, including save-state and overlap conditions, while the actuation path must deliver sufficient heater current without placing a high-power stage inside the low-voltage chip. This requires careful coordination

between the state controller, pair decoder, off-chip current driver, and MEMS heater branches.

Block	Team Member(s)
Circuit FSM State Control	Ruilin Yu, Michael Villeneuve
Circuit Pair Decoder	Ruilin Yu
Circuit Off-Chip Heater Driver	Ruilin Yu
MEMS Actuators	Elliot Measel
MEMS Spring	Steven Pereanu
MEMS Fabrication Details	Elliot Measel

Table 2. Team responsibilities

III. DESIGN AND SIMULATION

A. FSM Logic and Circuit (Ruilin & Michael)

The finite-state machine (FSM) controller defines the digital motion sequence for the bidirectional electrothermal inchworm actuator. It takes three global control inputs – CLK, enable, and reset – and produces nine one-hot state outputs S1 through S9, which are passed directly to the pair decoder to determine which heater branches are activated at each step.

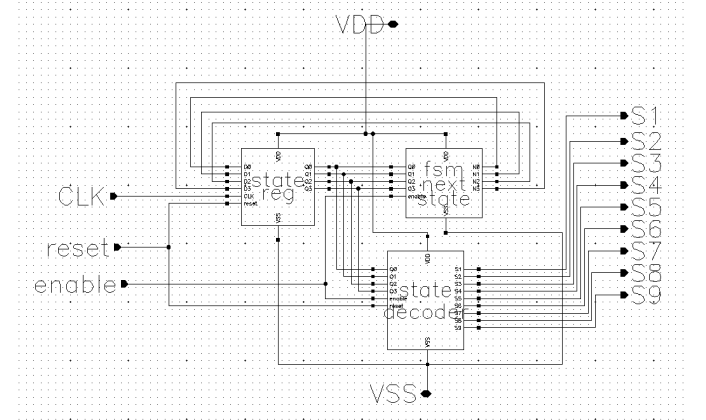


Fig. 7. Digital schematic of the FSM state, including the input and output identifications

Circuit Architecture. As shown in Figure 7, the FSM is composed of three sub-blocks. The *state_reg* block consists of four D flip-flop clocked on the rising edge of CLK, storing the current 4-bit binary state. Four bits are sufficient to encode all nine active states (S1-S9). The *fsm_next_state* block implements the transition logic: it increments the binary state by one on each valid clock edge when enable = 1, and wraps back to S1 after S9, forming a modulo-9 counter. When enable = 0, the counter holds its current value and no transition occurs. When reset = 1, the counter is forced asynchronously to the initial state regardless of the clock. The *state_decoder* block converts the 4-bit binary state from *state_reg* into nine mutually exclusive one-hot signals S1–S9, which serve as the

outputs of the FSM and the inputs to the pair decoder.

Timing design. The clock frequency of 33.33 Hz corresponds to a clock period of 30 ms, which sets the dwell time for each motion state. With nine active states (S1-S9), one complete inchworm cycle takes 270 ms. The state register requires four flip-flops to encode the nine states. Each D flip-flop is clocked on the rising edge, and the next-state logic implements a modulo-9 counter gated by the enable signal.

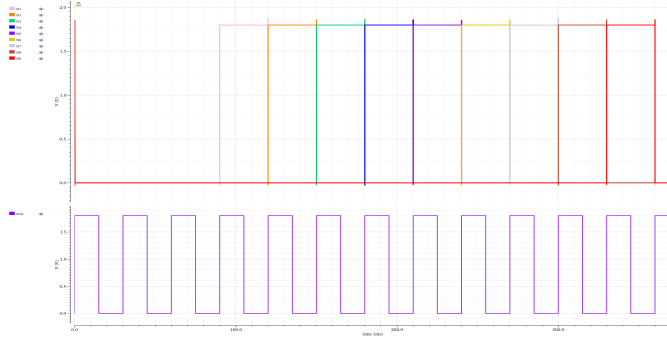


Fig. 8. Digital simulation of the finite state machine

Figure 8 shows the simulated waveforms of the FSM under normal operating conditions (enable = 1, reset = 0). The upper panel displays S1 through S9 as one-hot voltage signals at 1.8 V logic levels, each asserted for exactly 30 ms in sequence. The lower panel shows the 33.33 Hz CLK input. The simulation confirms that the FSM advances monotonically through $S1 \rightarrow S2 \rightarrow \dots \rightarrow S9$ without glitches or missing states, and that only one state signal is high at any given time. After S9 completes, the sequence wraps back to S1 for the next cycle, generating the continuous periodic activation pattern required for inchworm motion.

Overall, the finite-state controller serves as the timing and sequencing core of the circuit subsystem. It defines the ordered motion states required for the inchworm operation and assigns the correct time to each stage, providing a compact digital interface to the pair decoder and enabling robust branch-by-branch actuation in both motion directions.

B. Pair Decoder (Ruilin)

The pair decoder translates the one-hot FSM state outputs into heater branch activation signals. It receives nine one-hot state signals S1–S9 from the FSM, along with the global direction, enable, and reset inputs, and produces five output control lines: Pair1_enable, Pair2_enable, Pair3_enable, Pair4_enable, and SS_enable. These five signals are passed directly to the off-chip current driver, which uses them to select which MEMS heater branches receive current.

State-to-Heater Mapping. Table. 3 defines the complete mapping from FSM state to heater activation for both directions. The leftward sequence is the mirror image of the rightward sequence, so the same FSM progression supports bidirectional motion through a single direction control bit, without requiring a separate state machine for each direction.

State	Rightward Motion	Leftward Motion
S1	P1	P4
S2	SS + P1	SS + P4
S3	SS + P1 + P2	SS + P3 + P4
S4	SS + P2	SS + P3
S5	SS + P2 + P3	SS + P2 + P3
S6	SS + P3	SS + P2
S7	SS + P3 + P4	SS + P1 + P2
S8	SS + P4	SS + P1
S9	P4	P1

Table 3. State-to-heater mapping table

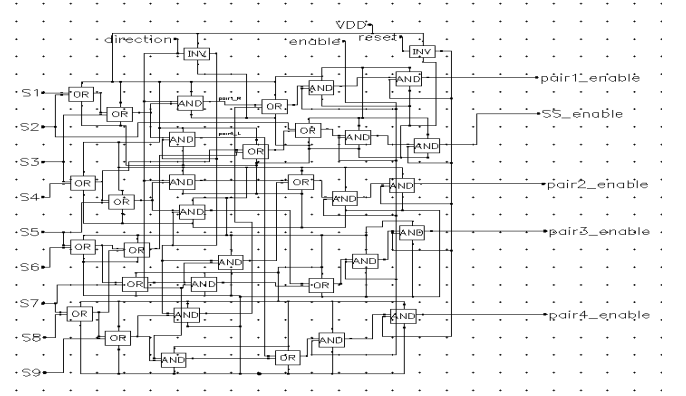


Fig. 9. Digital schematic of the pair decoder

Circuit Architecture. As shown in Figure 9, the order is organized into three combinational stages. In the first stage, OR trees compute the raw rightward and leftward activation conditions for each output. For example, the rightward condition for pair1_enable is asserted in states, S1, S2, and S3, giving the Boolean expression:

$$\text{pair1_enable(right)} = S1 + S2 + S3$$

and the save-state output is active across state S2 through S8:

$$\text{SS_enable} = S2 + S3 + S4 + S5 + S6 + S7 + S8$$

In the second stage, the direction bit selects between the rightward and leftward OR-tree outputs through AND-OR multiplexer logic. The non-inverted direction signal gates the rightward path, while the inverted direction signal gates the leftward path, and the two are OR-combined to form the pre-gated output. In the third stage, each output is ANDed with enable and with the complement of reset, so that all five outputs are forced low whenever the system is idle or being initialized. This three-stage structure is visible in Figure 9: the left half of the schematic contains the OR trees and direction multiplexers, while the right column of AND gates implements the final enable/reset gating before each output.

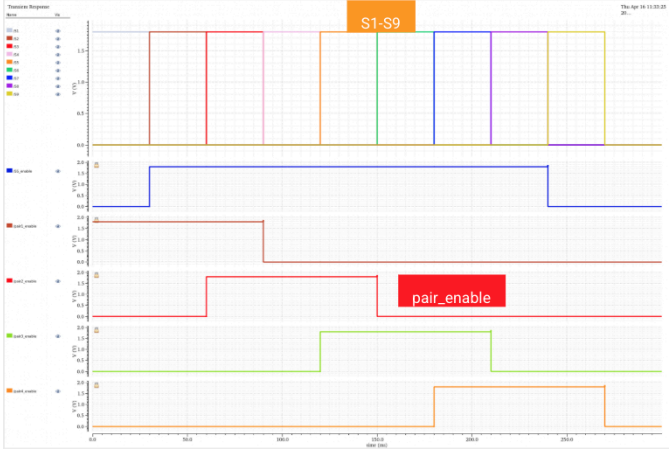


Fig. 10. Simulated pair decoder outputs for the case $S3 = 1$, $enable = 1$, and $reset = 0$. For rightward motion, the decoder produces $SS_enable = 1$, $Pair1_enable = 1$, and $Pair2_enable = 1$, while $Pair3_enable = 0$ and $Pair4_enable = 0$.

Simulation Results. Figure 10 shows the simulated transient waveforms of the pair decoder. The upper panel displays the nine FSM state inputs $S1-S9$ cycling sequentially, each held for 30 ms. The lower panels show the corresponding decoder outputs SS_enable , $Pair1_enable$, $Pair2_enable$, $Pair3_enable$, and $Pair4_enable$. For the rightward direction case shown, SS_enable rises at the start of $S2$ and remains high through $S8$, consistent with the mapping in Table 3. $Pair1_enable$ is high during $S1$, $S2$, and $S3$, then falls. $Pair2_enable$ overlaps with $Pair1_enable$ during $S3$, creating the simultaneous two-pair activation required for the handoff step. This overlap pattern repeats for each successive pair transition, confirming that the decoder correctly produces the staggered activation sequence needed for continuous inchworm motion. Non-active outputs remain at 0 V throughout, and all outputs drop to 0 V when enable is deasserted at the end of the simulation window.

Overall, the pair decoder serves as the key interface between the digital motion controller and the heater-driving stage. It converts discrete motion states into physically meaningful heater activation patterns, enabling synchronized save-state engagement and bidirectional pair-to-pair stepping.

C. Off-Chip Heater Driver (Ruilin)

Architecture and Design Motivation. The heater-driving architecture is implemented as a split system between the on-chip CMOS controller and an off-chip power stage. The CMOS chip produces only five low-voltage branch-selection signals — $Pair1_enable$, $Pair2_enable$, $Pair3_enable$, $Pair4_enable$, and SS_enable — received from the pair decoder. The actual heater current is delivered by an off-chip switching stage powered by an external 9 V supply.

This partition is motivated by the voltage and current mismatch between the CMOS process and the MEMS heater requirements. The 180 nm CMOS process operates at a nominal VDD of 1.8 V, while each heater branch requires approximately 9 V to deliver the target 100 mA. On-chip

integration would require thick-oxide high-voltage transistors and careful layout isolation, significantly increasing design complexity. Alternative approaches such as an on-chip high-side PMOS current source or a dedicated power management IC were considered but rejected for the same reason. The off-chip approach cleanly separates the digital control domain from the power delivery domain, allowing each to be designed, verified, and debugged independently.

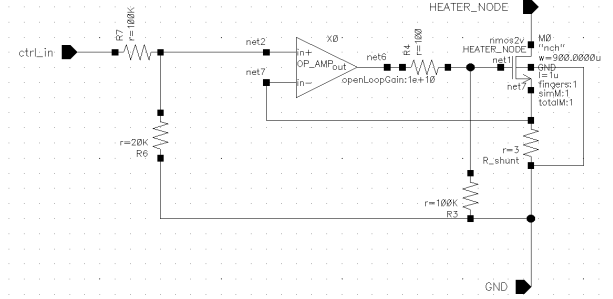


Fig. 11. Digital schematic of the off-chip current driver

Circuit Description. Figure 11 shows the schematic of a single heater driver channel. The control input $ctrl_in$ receives the $pair_enable$ signal from the CMOS chip. A voltage divider formed by $R7$ (100 kΩ) and $R6$ (20 kΩ) scales the reference level, setting the op-amp non-inverting input to:

$$V_{ref} = V_{ctrl_in} \times R6 / (R6 + R7) = 1.8 \times 20/120 = 0.3 \text{ V}$$

The op-amp drives the gate of the NMOS transistor in a feedback configuration, with R_shunt (3 Ω) as the sense element in the source path. The feedback forces $V_shunt = V_ref$, so the output current is regulated to:

$$I_{out} = V_{ref} / R_shunt = 0.3 / 3 = 100 \text{ mA}$$

$R3$ (100 kΩ) provides the gate return path. The NMOS connects the $HEATER_NODE$ to GND through R_shunt , completing the current path from the external 9 V heater supply through the MEMS heater branch.

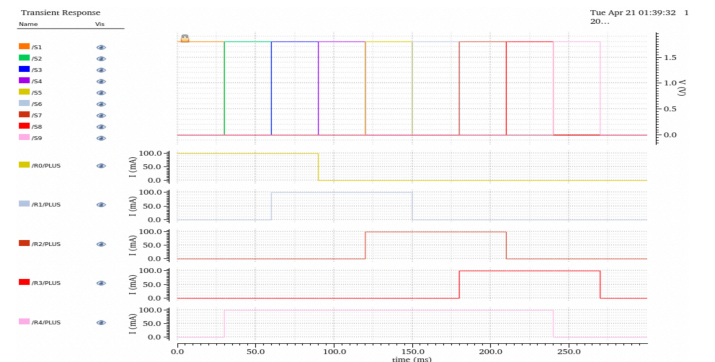


Fig. 12. Simulated heater branch currents for the case $S3=1$, $enable=1$, and $reset=0$. The system delivers approximately 100 mA to the SS, P1, and P2 branches, while the remaining branches remain off.

Simulation Results. Figure 12 shows the simulated branch

currents for all five heater channels across one complete inchworm cycle. The top panel shows the FSM state inputs S1–S9. The lower five panels show the currents in four actuators (represented by R0, R1, R2, R3, and R4), corresponding to the SS, Pair1, Pair2, Pair3, and Pair4 branches respectively. Each active branch receives approximately 100 mA during its assigned states, and drops cleanly to 0 mA when deactivated. The staggered overlap pattern — where two branches are simultaneously active during handoff states such as S3 (SS + P1 + P2) — is clearly visible in the R0, R1, and R2 panels between 50 ms and 100 ms. This confirms that the current driver correctly follows the pair decoder's enable signals and delivers the required current profile to drive the MEMS inchworm sequence.

D. MEMS Actuator Design and Simulation (Elliot)

The most important part of our MEMS device affecting our performance is the actuator design. Each actuator needs to provide ~ 8 microns of movement and up to ~ 12 mN of lateral force. There are many limitations dictating how well we can meet these requirements, including pad current limits, buckling propensity, and inchworm tooth size. The force requirement will vary based on how the spring structure is designed and fabricated (discussed in the next section), but the 8 micron displacement requirement is directly a result of the minimum system geometry.

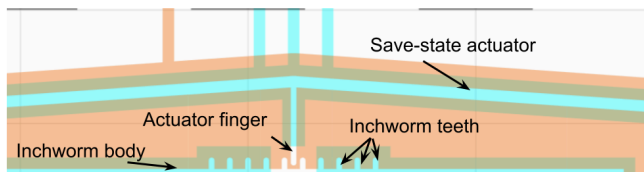


Fig. 13. Naming convention for our different design parts

Figure 13 above quickly reviews a few key terms regarding device interactions, and acts as a useful reference for this section of the report. It is also worth noting that the entire MEMS layout (with the exception of the spring) is perfectly mirrored from top to bottom. Every actuator on top has a flipped actuator on the bottom, which serves to cancel out and lateral forces on the spring, constricting its motion to the x-direction (left and right).

Inchworm Tooth Spacing and Geometry

For our MEMS fabrication, we were given a minimum feature size of 5 microns and a minimum gap size of 3 microns. First, we will discuss why we designed our teeth with such a specific width at a set distance apart. We planned to implement a “save-state” actuator on the top and bottom of our device, which when idle, rests in between the teeth of the central inchworm structure. This requires us fabricating this specific pair of actuators with the finger already placed in between the teeth. Figure 14 below shows the required dimensions clearly - in one period of actuation, we need to fit both the width of two 5 micron silicon beams (the actuator finger and the inchworm tooth) along with the two 3 micron

gaps between them. This results in a minimum actuation period of 16 microns for our system.

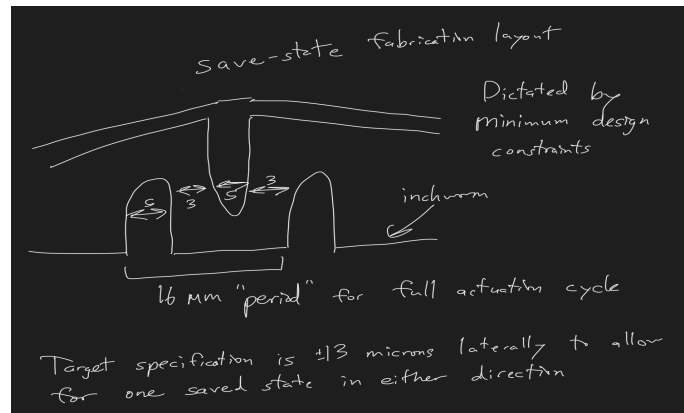


Fig. 14. Fabrication rules regarding silicon geometry dictate the period of placement for inchworm teeth

It's worth noting that our minimum target specification is not 16 microns, but only 13 microns. If we pull the save-state actuator up, move the inchworm to the right, and re-engage the save-state actuator, we only move 13 microns, since the spring on the left is applying tension to the inchworm. This means that, when moving to the right one period, we need the right side of the actuator finger to line up in contact with the left side of the next inchworm tooth, giving us our 13 micron target specification. Every period after this point would then require an additional 16 microns of movement.

Justification for Total Actuator Count

The 16 micron period also describes our requirement for having 10 actuators in total. The save-state actuator by itself can provide for up to 5 microns of displacement within the period - this would occur if we lined it up slightly off-center and pressed it down into the inchworm. We need to add an even number of actuators to cycle through the full period, since the inchworm is meant to have equivalent functionality moving to the left and right. If we add just two actuators, each one must account for 5.5 microns of bidirectional movement, rounding out the 16 micron period. To provide this displacement, placing the actuator just off center above the inchworm tooth and again pressing it down, the actuator must be 6 microns wide. Assuming this is the case, if the 6 micron actuator finger is engaged with the inchworm teeth, then there is a 5 micron gap of inchworm movement before it hits the next tooth. Since each stage provides 5.5 microns of movement, this would cause gridlock and break the system.

Running the thought experiment again with 5 total actuators proves feasible. If each actuator finger is the minimum width of 5 microns wide, each one can provide 3.2 microns of inchworm displacement without overcrowding the space in between inchworm teeth during tradeoff. So, we need 5 actuators in total to achieve periodic displacements, and accounting for the mirrored constraint discussed earlier, this comes out to 10 total actuators.

Tip Design for Contact Points

For all of these actuators, the shape of the tips for both the actuator finger and the inchworm teeth was discussed as a design consideration. A steeper angle, representing very sharp teeth, would require less output force from the actuator to move the inchworm, and would improve functionality in the event that silicon-silicon contacts provide high amounts of friction. A shallower angle would require less displacement from the actuator due to its higher mechanical advantage, and would also be fabricated more faithfully, since it doesn't have sharp edges.

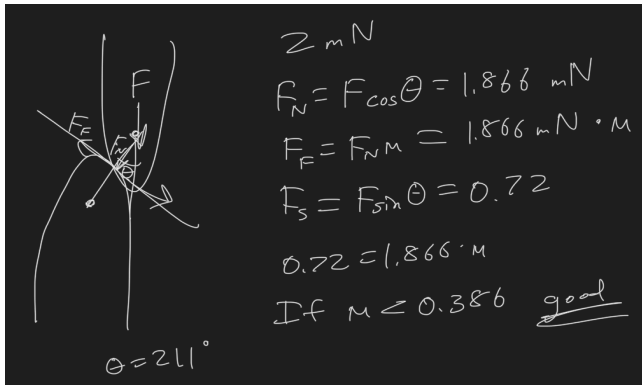


Fig. 15. Rounded tip contact will work for friction coefficient below 0.386

Both of these designs have their pros and cons, so we tried to combine the best of both worlds by rounding the tips. The minimum contact angle in an attempt to initiate sliding is 21.1 degrees (dictated by the 3.2 micron maximum offset for an actuator finger contacting an inchworm tooth). This indicates that as long as the coefficient of static friction between these two surfaces is less than 0.386, the surfaces will slide and create a pull-in effect, providing the full 3.2 micron displacement for that stage. There isn't much data for how much friction we can expect here, so this could limit our device's success. In preparation for this unknown parameter, we've designed an additional die to address this issue - to be addressed in later sections.

Actuator Vertical Displacement Requirement

Next, we will discuss how we arrived at the 8 micron vertical displacement requirement. Figure 16, shown below, helps illustrate why our actuators need to move roughly 8 microns to consider our device functional. The first 3 microns of vertical actuator movement are making up the 3 micron gap required by our fabrication rules. Since we chose rounded tip contacts, there is an additional radius of 2.5 microns on each tip to surpass before the thickest parts of the beams are in contact, ensuring we get the full displacement we desire.

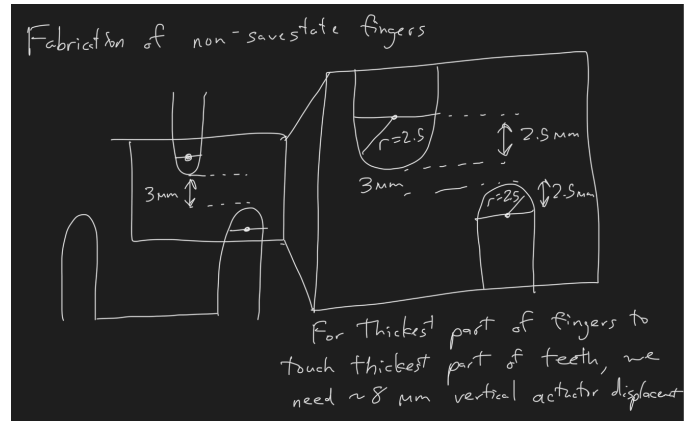


Fig. 16. Calculation associated with determining 8 micron displacement requirement for each actuator

The justification for the save-state actuator's required vertical displacement is a lot simpler. The inchworm teeth were somewhat arbitrarily chosen to be 10 microns tall each. We chose to embed the save-state actuator with 7 microns of vertical overlap from the inchworm teeth. Using a similar displacement requirement of 8 microns here allows this actuator to move sufficiently far out of the way for the inchworm to move left and right without interference.

Electrothermal Actuator Design

After calculating the target specifications for each one of these actuators, we started spec'ing out the specific actuator geometry to reach displacements of 8 microns. As a general rule of thumb, we chose to keep the maximum temperature of our actuator at 450° C or less during operation to prevent extended wear [6]. Length, the most important factor, was limited by how much horizontal space we were given to work with (~ 1800 microns). To keep space for the spring, power lines, and operating inchworm movement, we chose to make each actuator arm 600 microns long for a total length of 1200 microns between anchored ends.

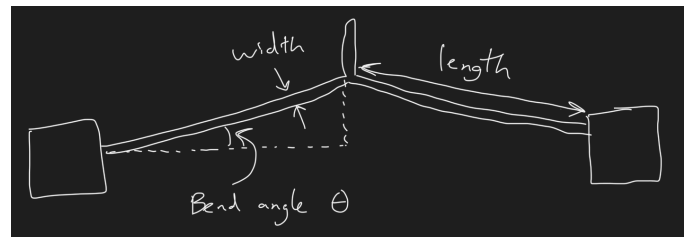


Fig. 17. Image of one actuator with length, width, and bend angle delineated

The next parameter we chose to optimize was the bend angle. Increasing the bend angle helped increase output force, with one downside - less output displacement. Using an initial actuator width of 10, we looked at output force as a function of bend angle, and chose our angle based on our force requirement from the spring design (at least ~ 12 microns).

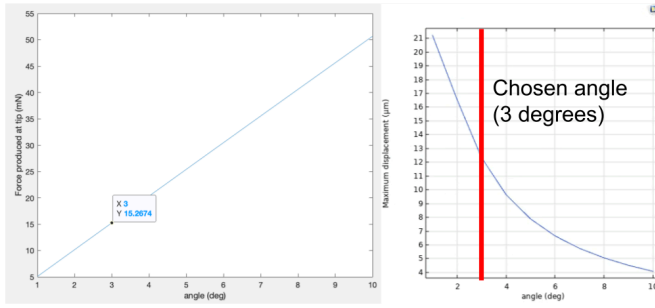


Fig. 18. Chosen bend angle of 3 degrees supplies sufficient force and displacement for our design requirements

Width was the last parameter to consider. Increasing width didn't have much effect on displacement, as long as we could scale power to keep the device at the same steady state temperature. The main tradeoff here was actually with force. Initial calculations and simulations indicated that a 10 micron wide actuator would be able to support force from 15-30 mN, sufficient for operating at the edge regions of our spring. Furthermore, to achieve this displacement, it took about 70 mA of current to operate, which is well below our design limitation of 200 mA per pad. Future improvements to this project would include more in-depth 2-dimensional parameter variation and analysis to avoid local minima, but for our purposes, the current setup seemed to prove feasible.

Around this point, we did some research on in-plane buckling of MEMS electrothermal actuators, and realized this could present a problem. After testing with our current design dimensions, we got conflicting results - our simulations showed that our device wouldn't buckle while our calculations showed that it would. We figured it could've been an issue with how COMSOL determines buckling loads, so to be safe, we increased the bend angle slightly 4 degrees to help overcome this. We didn't want to change the other parameters too drastically, as they were already specced out for power requirements, but we did include a more conservative, buckling resistant design in a separate die in case this turned out to be an issue.

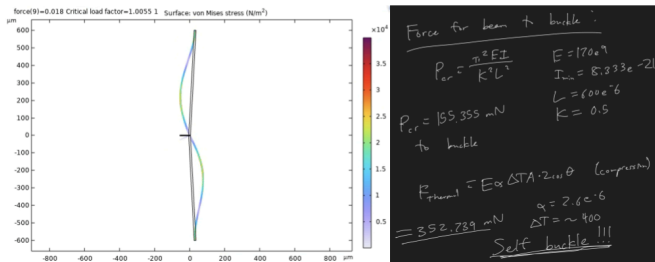


Fig. 19. Simulation says actuator buckles with ~18 mN applied force, calculation says actuator will self-buckle

Electrical Current Uncertainty

While we verified the previous sections would work within certain current ranges, it's hard to say the exact value for current we are sending. We weren't given any specific values for thermal coefficient of resistance or thermal conductivity

for our p++ doped silicon material, both of which affect the current requirements. We found a variety of different values online, and most tests showed that we needed around 55 mA to operate our devices, as opposed to the previously expected current of 70 mA.

We also briefly discussed the effect of air conduction and convection on heat transfer from our actuators to the environment. We were told that air can mostly be neglected, so this shouldn't be an issue, although some simulations indicated a required current closer to 80-90 mA. The specific value needed here will need to be gathered during the MEMS testing phase.

Since the current limit per pad is 200 mA, and each actuator is connected to one pad, we should be able to send up to 200 mA to each actuator to reach our desired steady state temperature. We failed to take one detail into account though - in an effort to reduce the number of pad connections required, we shorted the grounds of all the non-save-state actuators together. If any two of these actuators are operating at the same time, we can only send 100 mA to them. In a future iteration of this project, a simple workaround would've been to stagger all of the actuator stages between the two large ground connections instead of rippling them sequentially either inward or outward.

Secondary Die Specifications

As briefly mentioned in previous sections, we uncovered a few uncertainties in designing and simulating our primary die. We created a second die 2 identical to Die 1 (our primary die) with 3 exceptions. Firstly, it includes 2 spring to inchworm connections as opposed to 1, which is discussed in the following section. Secondly, it includes sharp, 45 degree angle teeth instead of rounded teeth. This allows the device to work assuming a static friction coefficient of 1, although it technically requires slightly more displacement out of each actuator. Lastly, the actuators have been widened from 10 microns to 15. Calculations now verify that this system won't buckle, and the only concern is getting enough displacement out of this larger actuator without exceeding the aforementioned power limits.

In addition to this secondary die, we also produced a more robust third die in case neither of our first two dies proved functional. This one is much simpler, involving only two stages of actuation, but still demonstrates cascaded multi-stage movement of a mass on a spring using physical contact.

E. MEMS Spring Design and Simulation (Steven)

Serpentine Spring Selection

The suspension spring was designed to provide a low lateral spring constant so that the inchworm body could achieve the target lateral displacement of 50 μm using the limited force available from the actuators. Because the actuators occupy a large portion of the available chip area, a serpentine spring was selected since it provides high lateral compliance while using relatively little horizontal space [7], [8]. In addition, the overall device is mirrored about the centerline, which reduces

parasitic moments and helps constrain the spring motion primarily to the x-direction.

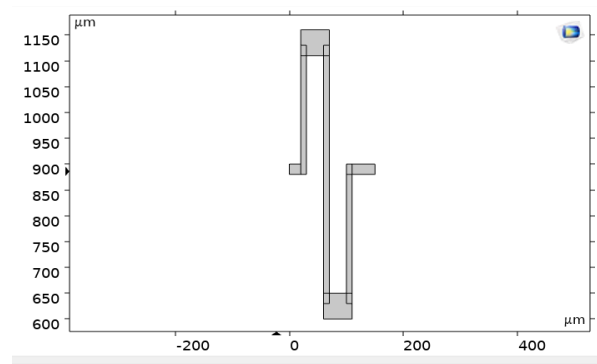


Fig. 20. A close up of our serpentine spring

Spring Design Requirements

From actuator calculations, the total available force was estimated to be approximately 15 mN. However, only a fraction of this force contributes to useful lateral motion of the inchworm body, and roughly one-third was assumed to act in the desired direction. Therefore, to obtain the target displacement of 50 μm , the suspension needed an effective spring constant of approximately $k \approx F/x \approx 100$ N/m. The final serpentine spring design achieved a simulated lateral spring constant of 87.8 N/m, while hand calculations predicted 96 N/m. This agreement indicates that the spring meets the displacement requirement and that the analytical model captures the first-order mechanical behavior reasonably well.

Fabrication Constraints and Beam Sizing

To balance compliance with fabrication robustness, the spring beam width was set to 10 μm . This is twice the minimum 5 μm silicon feature width and was chosen to improve fabrication yield while maintaining a sufficiently low spring constant. Matching the spring beam width to the actuator beam width should also help produce more consistent DRIE behavior across the device. Although a lower spring constant could have been obtained with narrower beams, that approach would have made the structure more fragile and more sensitive to fabrication variation. This is especially relevant because our actuator beams are 10 μm thick and having consistent beam thickness helps in fabrication.

Out of Plane Stability and Buckling

The device has only one spring on the side so gravity will cause the entire thing to droop downwards as shown in the figure. Out of plane behavior was evaluated to confirm that the serpentine suspension would not introduce excessive out of plane motion. The predicted out of plane displacement of the inchworm body was only 0.05 μm in simulation and 0.155 μm by calculation, indicating that the structure should retain sufficient clearance from the glass substrate during operation. However after seeing the devices manufactured there seems to be a significant drop that is causing some testing irregularities.



Fig. 21. Die 2's spring design vs die 1's spring design

Two Spring Variants for Dies 2 and 3

For dies 2 and 3, a two spring suspension was introduced as a design variation to increase out of plane stiffness and reduce the risk of tilt during operation. Although the baseline calculations predicted only small out of plane droop, the two spring configuration provides additional constraint and distributes the load more evenly across the moving body. To preserve the required low lateral spring constant, each spring in this version had to be made narrower, which introduced a fabrication tradeoff because the spring dimensions no longer matched the actuator beams as closely. These secondary dies were therefore included as an experimental comparison to determine whether the improvement in out of plane stability outweighs the possible reduction in fabrication consistency.

F. MEMS Fabrication Details (Elliot)

Fabrication Process

The fabrication process to be used for this device was given to us directly by course staff. We were allowed to make four total masks - one for the glass recess, one for the bottom electrode, one for the aluminum pads, and one for the silicon (DRIE). The process is shown in detail below.

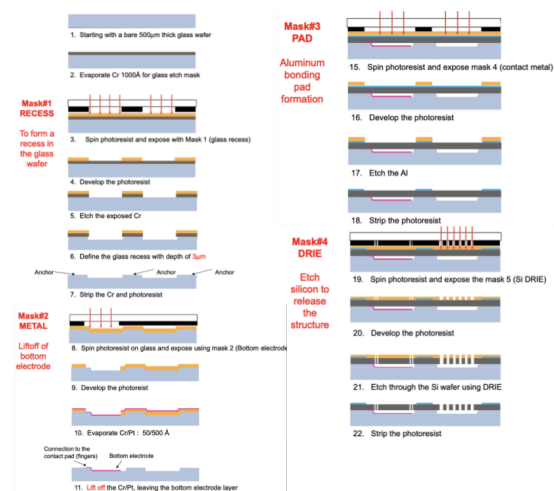


Fig. 22. MEMS fabrication procedure directly from lecture slides, steps 12-14 (missing) detail bonding of silicon wafer onto glass and deposition of Al

To summarize the process for making our MEMS device, we first took a bare 500 micron thick glass wafer, patterned

and etched a recess into the glass, patterned and deposited platinum onto the glass, and bonded a 100 micron silicon wafer on top. Then, we patterned aluminum on top, and lastly used DRIE to etch the silicon wafer all the way down to the glass.

General Layout Rules

When designing the final layout for our device, we had to adhere to a set of design rule constraints for layout generation. For the final silicon structures, minimum feature size is 5 microns and minimum gap size is 3 microns. We were also required to connect our entire glass recess to prevent the creation of a pressure chamber, which would break the device during fabrication. Most other direct DRC rules, such as aluminum and platinum pad feature sizes, didn't affect our design that much. But, there were additional guidelines to go along with these rules that were tried to adhere to. It was suggested to keep the total length of any suspended thin silicon beams to 1mm or less, to avoid overheating during the DRIE process. Our spring was right around this value, so we thickened it up to help dissipate heat faster. We were also recommended to provide aluminum pads around the edge in the same configuration as the existing probe card, so that the MEMS testing phase would be more straightforward.

We defined a few general guidelines on our own, too. For silicon structures that carry electrical signals (voltage or current) we have a minimum gap size of 10 microns and a minimum feature size of 30 microns (excluding the actuators). For silicon structures that involve movement, our minimum gap size is 20 microns. Our entire die (with the exception of the spring in die one and the aluminum layer) is mirrored vertically - the top and bottom half should behave the same. Large open gaps were filled with silicon anchored to glass, and the areas around the actuators were kept as symmetric as possible to minimize the lateral drift during operation.

Die 1 Layout

The layout for die 1 is shown in Figure 23. To briefly recap how this die works, the central inchworm structure is connected to a spring from the left side of the die. The movement of the inchworm is restricted to the x-direction (left and right) and is actuated by the pinching motion of electrothermal actuator pairs. These pairs will operate in a specific order to either move the whole structure to the left or right, and the save-state actuators at the top and bottom will be able to hold the spring in tension or compression, even after the device is powered off.

Initial designs included a spring on the right side to anchor the inchworm structure from both sides of the die, creating a symmetric design both vertically and horizontally. The plan was to power the actuators using platinum pads deposited on the glass layer, but we quickly realized that these wouldn't support the power requirements needed for high current electrothermal actuators. We ditched the spring on the right side and created thick power line connections to deliver current to the actuators.

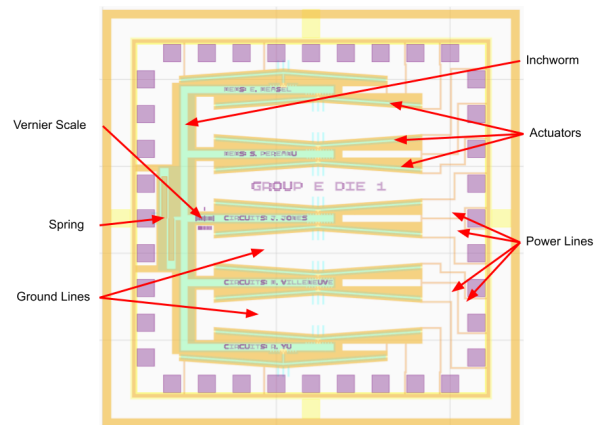


Fig. 23. MEMS Layout Die 1. Yellow - glass recess, blue - platinum electrode, orange - DRIE mask, purple - aluminum pads

The glass is recessed beneath all moving parts of our device. This includes the spring, the inchworm, and the actuators. A platinum electrode was also added beneath these parts. This was done to minimize back etch on our very meticulously designed mechanical assemblies, and we don't plan on using these pads for any electrical purposes. We didn't place pads directly under the contact point between the actuator fingers and inchworm teeth, as this risked creating an electrical short, potentially breaking our device upon testing.

We have set up a vernier scale within the aluminum pad mask layer, which will help us measure exactly how much displacement our inchworm is getting. The rest of the aluminum pad layer consists of the electrical pads around the edge and writing with our names and die description.

IV. TEST RESULTS - MEMS (ELLIOT)

Upon finishing the design, simulation, and layout of our MEMS device, we were able to fabricate it in house. We were given three separate 3-hour test sessions to characterize our device and its performance compared to simulation and calculation.

In general, our devices didn't turn out to be as functional as we had hoped. Die 1 and 2 had significant issues initiating a sliding surface at the contact interface (although die 2 did show a few promising results), and the reasoning will be discussed in upcoming sections. Die 3 was very consistent in operation, proving that our proposed operation principle is still very possible and maybe even realistic for industrial applications, just that it requires more fine tuning for future design iterations. Table 4 below shows a summary of expected vs measured results for a few key parameters.

Specification	Expected value	Measured value
Target Minimum Displacement	+/- 13 μm lateral movement	+/- 3 μm
Actuator Current (to reach 8 μm displacement)	100 mA	95 mA
Die 1 Median Actuator Resistance	60 Ω	38.168 Ω
Die 2 Median Actuator Resistance	40 Ω	25.406 Ω

Table. 4. Calculated values in design phase compared with measured values on MEMS device

MEMS Test Procedure

For all of our test results, we conducted experiments in the same way to help isolate the other parameters we were looking at analyzing. Dies 1 and 2 were very similar, so we used the same base test procedure for both. We started by finding the resistance values for all 10 actuators by applying 1 mA of current one at a time and measuring the voltage. We then verified that they were within a magnitude of the expected values (60 Ω for die 1, 40 Ω for die 2). More importantly, this told us that there weren't any short or open circuits in unwanted places, meaning it was safe to test.

Once we verified the resistance values, we began testing the actuators. We first checked the minimum current required to pull the save-state actuators out from the teeth. We tested both of these actuators in parallel, starting with 50 mA and incrementing by 10 mA until we saw clearance. There wasn't an easy way to verify this for sure, since we were relying on visual feedback, but we did the best with the camera software and made sure that at no time we would get physical collisions that we weren't planning on. Pair one was measured the same way, and once we had the required currents for these two stages, we moved towards 3 stage operation.

We used 3 current sources for this section - 1 for the save-state actuators, 1 for pair 1, and 1 for pair 2. Using the known currents required for actuation of the save-state and pair 1 actuators, we would trigger the current sources in the following order: SS -> Pair1 -> Pair2 -> Pair2 -> Pair1 -> SS. For pair 2's current, we again started with 50 mA and worked our way up until we saw contact, still never to exceed the 200 mA limit for pair1 and pair2 combined.

For die 3 tests, we underwent a very similar procedure, first measuring the resistance values of all 4 actuators. Then, we would send short pulses to pair 1 starting at 50 mA and increasing by 10 mA until we saw contact. Once this was the case, we would set up a second current source with 50 mA (again, increasing by 10 mA each trial until pair 2 made contact) and operate the system using the following order:

Pair1 -> Pair2 -> Pair2 -> Pair1. The current limit wasn't going to become an issue for this die, since each actuator had more than one pad for each terminal.

Specific Testing Results

The first major discrepancy we noticed between our predicted and actual device performance was with our resistance values. We expected die 1 actuators to have resistance values close to 60 Ω , and die 2 actuators to have resistance values close to 40 Ω . The actual values for all of our resistors are shown below - they were generally low by 30-40%. We figured the silicon doping concentration may have been significantly different than in our simulations. This is problematic for our devices, since lower resistance on our electrothermal actuators meant we needed to supply more current to reach the same steady state temperature. This wasn't a problem on die 1, and we were able to produce our target displacement of 8 μm using only 95 mA of current, slightly under our 100 mA limit. However, for die 2, adjacent actuators weren't able to be operated fully at the same time. It took 115-125 mA of current (depending on the actuator) to move the tips 8 μm , so if two actuators sharing a ground line were powered at the same time, one aluminum pad would see up to 250 mA of current. During testing, we would never exceed this threshold so as to not break the device or testing setup, and this meant that we couldn't test die 2 optimally.

Resistance Parameter	Die 1 (center)	Die 1 (edge)	Die 2 (center)	Die 2 (edge)
Mean (Ω)	49.482	35.014	25.944	27.092
Standard Deviation (Ω)	12.272	2.043	2.140	5.195
Count (#)	10	10	20	30

Table. 5. Resistances for die 1 and die 2 were much lower than expected

We ran into issues initiating a sliding interface between the actuator arms and the inchworm teeth. Instead of redirecting the vertical motion to the side to create lateral movement, the actuators simply "squeezed" the inchworm arms away. This phenomenon is shown below in Figure 24.

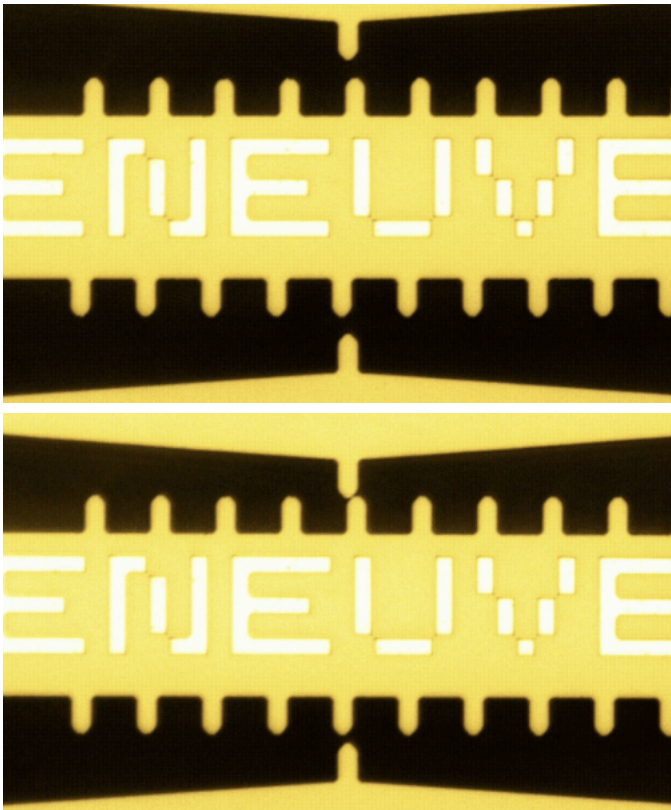


Fig. 24. No sliding interface created on die 1, resulting in vertical displacement of inchworm, pushing inchworm closer to actuator on bottom

We determined that there were two main issues causing this behavior. Firstly, the coefficient of friction at the silicon-silicon interface was too high to create a sliding interface, getting stuck upon contact instead. Die 2 proved that this may have been the issue, as we saw some success when using the sharpened 45° inchworm teeth (Figure 25). Here, we were able to redirect the force laterally and generate $\sim 3\mu\text{m}$ of displacement on the inchworm just through physical contact. This case shows that our system is viable as a method in general, but just needs fine tuning for specific parameters to make it more robust.

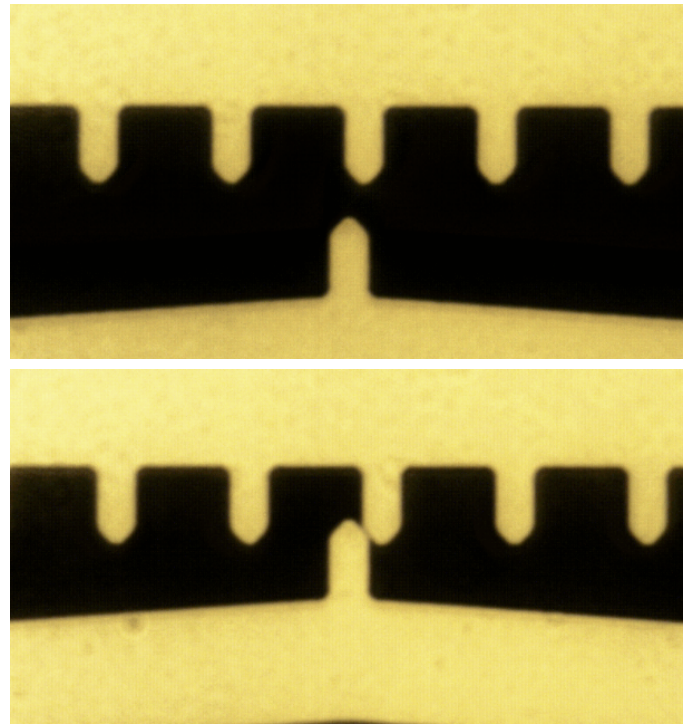


Fig. 25. Sliding contact initiated on die 2, moving inchworm $\sim 3\mu\text{m}$ to the right, evident in bottom frame

Secondly, the “squeeze” spring constant (from the connecting structure on the left of the inchworm to the contact point) for these inchworm arms was likely too low. If we increased the stiffness of the inchworm, it would be much harder for the actuators to push these arms around in relation to the rest of the inchworm structure, and we would see a stronger interface at the point of contact, capable of redirecting forces.

In an effort to supply more force at this point of contact, we tested pair 2 alone with increasing currents, pushing the inchworm arm further and further until it actually made contact with the actuator above it. Shown below is the broken actuator as a result of this test. From this test result, and using a quick calculation involving cantilever beam deformation, we determined that the thin actuators from Die 1 produced 2 mN of output force right before failure. This disproved our last minute thoughts before design submission that our actuators would self-buckle, and tells us that we have more room to thin out our actuators than we originally thought.

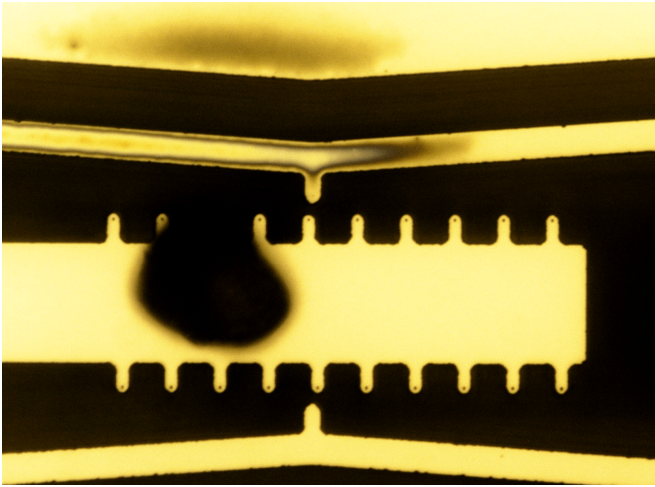


Fig. 26. Failure as a result of applying too much current to bottom actuator, pushing middle structure into contact with top actuator

We noticed a few other phenomena that could've marginally affected our device's performance. It appeared that the inchworm structure would start to move before we noticed visible contact from the actuator tips. This means that the bottom of the inchworm structure was likely initiating the contact, due to gravitational deflections after the silicon was released in fabrication. We also noticed that some of our actuators moved slightly even when we weren't sending currents directly to them. This can only be explained by parasitic capacitance values evident in the large power line structures on the right. Through standard device operation, we unintentionally apply 4 volts across 10 μm gaps in many places, likely contributing to some of these leakage issues.

Proposed Changes

If we were to design this device again, we would take into account a few key changes. Major examples include staggering the pair control to never reach our pad current limit, and increasing the spring constant of the inchworm arms to help direct forces more consistently. We would also increase the gaps between power lines connected to the actuators, minimizing the possibility of leakage current.

V. CIRCUIT TESTING STRATEGY AND PLAN (RUILIN)

The circuit testing strategy is organized into three levels of increasing integration: digital logic validation, current driver validation, and full chip-level validation. At each level, the expected outcomes are defined prior to testing based on simulation results, and a structured debug procedure is prepared to isolate failures to a specific block. The primary instruments used are a digital oscilloscope, a function generator for clock and control inputs, a DC power supply for VDD and V_HEAT, and a digital multimeter for resistance and DC voltage measurements.

Digital Logic Validation

The first stage verifies the FSM and pair decoder independently of the driver. The chip is powered at 1.8 V, with a 33.33 Hz clock applied to the CLK pad and DC logic levels

applied to enable, reset, and direction. The five decoder outputs — SS_enable and Pair1_enable through Pair4_enable — are observed with an oscilloscope. The expected result is the simulated overlap sequence, with each state lasting about 30 ms and output levels switching cleanly between 0V and 1.8V. Reset should force all outputs low, enable should pause the sequence, and direction should reverse the pair order. If the outputs do not match simulation, debugging begins by checking VDD, clock, and reset, then probing internal FSM state signals if accessible to determine whether the issue is in the FSM or the pair decoder.

Current Driver Validation

The second stage validates the off-chip current driver using a 70 Ω resistor as a substitute MEMS heater load, connected between HEATER_NODE and the external 9 V supply. One pair_enable output from the CMOS chip drives one driver channel. The output current is measured indirectly by probing V_shunt across R_shunt with a digital oscilloscope:

$$I_{\text{out}} = V_{\text{shunt}} / R_{\text{shunt}}$$

The expected result is $V_{\text{shunt}} = 0.3 \text{ V}$, corresponding to $I_{\text{out}} = 100 \text{ mA}$. If the measured current is lower than expected, the main checks are the op-amp output level, the divider resistor values $R_6 = 20 \text{ k}\Omega$ and $R_7 = 100 \text{ k}\Omega$, and the gate drive voltage at the NMOS pin. If the current is higher than expected, the likely cause is that the actual heater resistance is lower than the nominal 70 Ω , consistent with the 30–40% resistance shortfall observed during MEMS testing; in this case V_HEAT should be reduced proportionally to keep the branch current within the 100 mA limit. If no current flows at all, confirm that the 9 V supply is connected and that the ground of the driver circuit is tied to the same node as the MEMS heater ground pad.

Chip-Level Integration Validation

The final stage connects the verified CMOS chip outputs to the off-chip driver gates and the driver outputs to the actual MEMS heater pads. A digital oscilloscope simultaneously monitors the pair_enable signals at the chip output pads and the branch currents via the R_shunt voltages. The clock is applied at 33.33 Hz to enable high and direction set for rightward motion. The expected behavior is that each branch receives approximately 100 mA at the correct time step, with overlap states — where SS and one or two pair branches are simultaneously active — visible as multiple shunt voltages rising together. MEMS displacement is observed under a microscope and cross-referenced with the vernier scale on the MEMS die to confirm that electrical activation produces physical motion.

If the measured behavior does not match simulation, debugging proceeds in the following order. First, verify all supply and ground connections, ensuring the CMOS chip ground, driver circuit ground, and MEMS heater ground pad share a common node. Second, if unintended branches activate, inspect the MEMS power line routing for parasitic

capacitive coupling, which was observed during testing to cause leakage current across the 10 μm gaps between adjacent power lines at 4 V. Third, if timing errors or state skipping are observed, reduce the clock frequency below 33.33 Hz to provide additional margin for pad and wiring parasitics, then increase it incrementally until correct behavior is confirmed.

VI. SUMMARY

The proposed microsystem is designed to achieve large bidirectional displacement while maintaining low steady-state power consumption. The MEMS device uses electrothermal actuator branches to generate thermal expansion and transfer the resulting motion to central inchworm structure through staged tooth engagement. By coordinating multiple actuator pairs, the system intended to move the inchworm body to the left or right in controlled increments. Save-state actuators are incorporated to latch the inchworm in place and preserve the current displacement when the system is unpowered.

Expected performance for our MEMS device includes save-state displacements from 13 to 49 microns with 16 micron resolution when unpowered (in either direction). When powered, the system provides displacements from 0.2 to 49 microns with a resolution of 3.2 microns (also in either direction). We expect a maximum current draw of 100 mA, to be verified during testing, which would require up to 6 W of max power during certain switching phases - steady state power draw is still expected to be 0 when utilizing the save-state actuators. The maximum temperature reached by our actuators is 450° C, although we expect the system to be able to operate slightly below this temperature. We weren't able to reach our target minimum displacement to prove save-state functionality, but physical contact was proven to be a viable method for actuating a mass on a spring.

The circuit subsystem provides the electrical control path for the bidirectional inchworm actuator. A finite-state controller generates the ordered motion states, and a pair decoder converts those states into five heater-enable outputs corresponding to the four actuator pairs and the save-state branch. Together, these blocks produce the overlap sequence required for staged bidirectional motion while supporting global enable, reset, and direction control. At the system level, the top-level simulation verifies that the digital control path generates the intended heater-activation pattern and correctly coordinates the branch handoff sequence.

To satisfy the MEMS heater requirement without integrating a high-power stage into the 1.8 V CMOS chip, the heater-driving architecture was implemented as an off-chip current-regulated driver. The control signal from the CMOS chip is converted into a reference level through a resistor divider, and an op-amp feedback loop regulates the branch current through shunt sensing. This means the driver targets approximately 100 mA for each active heater branch while leaving the sequencing logic on-chip. Overall, the circuit design demonstrates that low-voltage CMOS control can be effectively combined with an external current-driving stage to realize the timed actuation sequence required by the bidirectional electrothermal inchworm system.

Category	Specification	Expected performance
MEMS	Target Minimum Displacement	+/- 13 microns lateral movement
	Maximum Displacement	+/- 50 microns lateral movement
	Displacement Resolution (powered)	3.2 microns
	Displacement Resolution (unpowered)	16 microns
	Actuator Current	50-100 mA
	Max Power Required	3-6 W
	Temperature Reached	450° C
Circuit	Digital voltage VDD	1.8V
	Heater supply VHEAT	9V
	Clock frequency	33.33Hz

Table 6. Final specification summary

Team contributions are listed in Table 7 below. If we were to design this MEMS device again, we would seek out more involved physical simulation software. COMSOL had limitations when trying to test electrical, thermal, and physical properties of contacts all at once, giving us much uncertainty in some areas. Specific design changes include thickening the inchworm arms to prevent them from getting squeezed, and staggering the actuator pairs to never overload one ground line and allow for higher power control.

Team Member(s)	Contributions
Elliot Measel	Fully designed MEMS actuators and die layout
Steven Pereanu	Simulated asymmetry with power lines and its effect on electrothermal actuator lateral movement
Jeovani Jones	Designed layouts for nand gates nor gates, and gates, decoder layout to pass LVS, and DRC
Michael Villeneuve	Designed circuit block diagram and layout for pair decoder
Ruilin Yu	Fully designed circuit architecture, developed all circuit schematics and FSM layouts, full top-level layout

Table 7. Team contributions to design process

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TEAM CONTRIBUTIONS

Elliot Measel

I submitted the idea for the design problem, and completed all of the early stage specifications to ensure the device had a possibility of working. I designed, simulated, and integrated an extensive part of the MEMS device - overseeing tests for everything except for the serpentine spring, and updating the design wherever necessary. I conducted much of the MEMS tests, and helped use our data to analyze and propose changes for future design iterations. For the presentations, I created and revised most of the MEMS slides. For the report, I completed the sections labeled with my name, and contributed a good amount to all other non-circuit specific sections.

Steven Pereanu

I designed the spring for the MEMS device. I also did various testing for MEMS device like calculating the spring constant of the arms or heat simulations to confirm there was no lateral movement in the electrothermal actuator. For the slides I helped make many of the MEMS slides. While for the report I did all the sections with my name, looked over the other MEMS sections and write a lot of the introduction.

Jeovani Jones

I submitted the system level modeling of the pair decoder as well as the general overall structure of the FSM. Throughout the project, I moved from working on the circuit block

diagram, to helping design the FSM to building out the full layout for the pair decoder as well as the layout for the NAND gate. I then performed DRC and LVS on both the NAND gate and well as the pair decoder making sure that both of them are functionally correct. For the presentations I revised and worked on circuit related slides most notably ones about the FSM and pair decoder.

Michael Villeneuve

I submitted/updated all circuit block diagrams throughout our project, and helped design the layout for the pair decoder, as well as the NOR gates/inverters that occupy it. I helped ensure that all aspects of our NOR gate and inverter layouts passed both DRC and LVS testing. I also designed a schematic of our FSM in Cadence that did not end up getting used in favor of a more applicable design. For the report, I helped complete all sections with my name attached, as well as revised the overall presentation, verifying all tables, figures, tables, and units and were both accurate and standardized throughout the whole report.

Ruilin Yu

I developed the overall circuit architecture for the microsystem and defined the detailed functionality of each circuit block. I completed the system-level modeling in Simulink, building a behavioral model to verify that the full control sequence and system integration would function as intended. I designed all circuit schematics in Cadence, including the FSM, pair decoder, and off-chip current driver. I completed the FSM block layouts, performed LVS verification for the pair decoder, and integrated the full top-level layout while completing both DRC and LVS checks. For presentations, I created and revised all circuit-related slides except for the circuit block diagram. For the report, I completed the sections labeled with my name and contributed to the overall circuit-system integration content.

GROUP COMMENTS

This course taught a lot about new and exciting topics in microsystem design and operation, but it was often hard to tell what was expected from us and where we stood from the instructor's perspective in relation to the other groups. It would've been nice to have feedback on past assignments, weekly meetings, and/or reports to help guide us in writing our final report, and tell us which areas we need to spend more time on in a general design sense.